

Farès Zaghouane

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SUMMARY

My work sits between representation learning and dynamical systems. At Apple I formalized hardware behavior as reachable-state specifications; at Nokia Bell Labs I built latent video-prediction models for cloud gaming and encountered a concrete stability problem: predictions drift once the semantic anchor becomes sparse. That is the failure mode I want to study: how much structure the predictor in a JEPA-style world model requires to remain coherent over long horizons.

RESEARCH INTERESTS

Keywords: World models · JEPA predictors · latent dynamical systems · state-space models · Koopman operators · energy-based models · latent geometry · representation learning · planning and control.

In joint-embedding predictive architectures, the encoder learns what to represent while the predictor models how representations evolve; that predictor is typically an unconstrained network. I tested whether explicit dynamical structure helps on a system where the prior is exactly correct after a change of basis. With the representation held fixed, the structured predictor did not outperform generic baselines; an oracle-basis experiment traced part of the penalty to the coordinates in which the prior was imposed rather than to the transition family itself. What I want to build next is a diagnostic framework that reveals when structure pays: which modes persist, what spectral margin precedes instability, and which latent directions actions can reach.

EDUCATION

École Polytechnique Fédérale de Lausanne (EPFL)

Lausanne, Switzerland

MSc Computer Science, Machine Learning specialization

Sept 2023 – Present

Relevant coursework: Artificial Neural Networks & Reinforcement Learning, Computer Vision, Machine Learning, Natural Language Processing, ML in Finance, Cryptography & Security.

CPE Lyon

Lyon, France

Diplôme d'Ingénieur, Computer Science — Software Engineering & Big Data track

Sept 2021 – Present

Relevant coursework: Software architecture, Design patterns, Databases, Data mining, Embedded systems, System administration, Quantum computing.

CPE Lyon

Lyon, France

Licence (BSc), Computer Science & Electronics

Sept 2020 – Sept 2021

Relevant coursework: Linear algebra, Algorithms, Signal processing, Digital & analog electronics, Discrete & continuous mathematics, Quantum mechanics.

EXPERIENCE

Nokia Bell Labs

Stuttgart, Germany

Research Intern (MSc Thesis)

Aug 2025 – Feb 2026

- The goal was to predict dense focus maps from cloud-gaming video for bandwidth and quality allocation under strict per-frame latency constraints, using a semantic signal as teacher and a lightweight model for deployment.
- Built a teacher–student video-prediction pipeline combining Grounding DINO semantic anchoring, RAFT optical-flow propagation, and a lightweight correction network to produce dense focus maps.
- Ran experiments with JEPA-style latent prediction, state-space models, and Mamba-Vision encoders; settled on a hybrid that met the real-time latency budget while matching semantic-only accuracy closely.
- Designed the pseudo-supervision setup, tracked temporal drift across pipeline modes, and built the dense-map evaluation protocol; the hybrid recovered most of the semantic-only quality at a fraction of the cost and outperformed motion-only and saliency-only baselines.

Apple

Munich, Germany

Design Verification Research Intern

Oct 2022 – Jul 2023

- Modeled IP behavior using finite-state machines and event-based abstractions, allowing long-running processes to be reasoned about without expanding every low-level timing step.
- Applied model checking to behaviors that simulation cannot cover exhaustively.

- Built Python tooling that generates Verilog/SystemVerilog assertions from high-level specs and validated it on industrial designs; FSM-level specs feed directly into the model-checker.
- The event-based abstraction set the level at which reachable states are specified: model checking operates there rather than at the cycle level, which changed which properties were practical to verify.

PROJECTS

Structured Latent Dynamics

Pre-registered factorial study of structured dynamical priors in the latent predictor of a JEPA world model: a Koopman-style family of 2×2 rotation-scaling blocks against MLP, transformer and selective-SSM baselines on coupled oscillators, where the generator belongs to the structured family after an appropriate change of basis. Arms are paired within backbone, so they share seed, data, rate and budget. So far structure does not buy rollout stability, on two backbones independently. The optimum is inside the class, so the transition family is not what fails; the diagnosis is that a structural prior of the Koopman family is a statement about a basis, not about coordinates, and a pre-registered oracle arm traces part of the penalty to that basis without reversing the sign.

Constrained Trace EBM

Energy-based verification of state-machine execution traces against temporal specifications, with an exact symbolic verifier as ground truth and the decision rules frozen before training. Detection on machines never seen in training holds at a bootstrap lower bound of 0.772 against a floor of 0.75 fixed in advance, a margin of 0.022; the remaining pre-registered claims are open and published as such; the study is in progress.

GPT-2 from Scratch

PyTorch implementation of a GPT-2-style decoder-only Transformer, built against the original papers: tokenization, token and positional embeddings, causal and multi-head attention, transformer blocks, pretraining, generation, and fine-tuning.

Machine Learning from Scratch

From-scratch ML curriculum: 52 course-style notebooks and a 123-module scikit-learn-style library. Each chapter takes one model apart rather than only re-implementing it: every equation term, every hyperparameter, and why one design supersedes the last. Runs from a custom autograd engine and NumPy estimators through Transformers, SSMs, GNNs and generative models to the geometry of representations — Riemannian optimization, optimal transport, equivariance.

RL from Scratch

From-scratch PyTorch RL curriculum: 25 algorithms from tabular Q-learning to PPO/SAC, Dreamer, MuZero, and an action-conditioned JEPA world model inspired by V-JEPA 2-AC, each taught in a course-style notebook then refactored onto a shared runtime. HalfCheetah-v5: SAC/TD3 $\approx 10k$, DDPG $\approx 11k$ near 1M steps.

V-JEPA2 ROI Semantic Extraction

Research prototype that organizes V-JEPA-style predictive video representations around semantic regions rather than uniform spatial patches.

VLM Data Engine

Video data pipeline covering shard processing, FPS conversion, scene detection, clip extraction, annotation, and quality checks; built to support multimodal world-model experiments.

SKILLS

ML & representation learning: PyTorch, TensorFlow, Neural networks, Self-supervised learning, JEPA / world models, Energy-based models

Vision & multimodal systems: Computer vision, Video understanding, Grounding DINO, RAFT / optical flow, Focus-map prediction, Multimodal data pipelines

Dynamics, control & RL: Reinforcement learning, State-space models, Mamba, Koopman operators, Dynamical systems

Mathematical & scientific foundations: Linear algebra, Applied statistics, Signal processing, Optimization, Spectral methods, Functional analysis

Software & formal methods: Python, NumPy, Jupyter, Formal verification, Model checking, Finite-state machines, Verilog / SystemVerilog assertions